Available online at www.sciencedirect.com

ScienceDirect

Journal homepage: www.elsevier.com/locate/cortex

Viewpoint

# Performance-based alternatives to race-norms in neuropsychological assessment

*Philip G. Gasquoine* [1]

*Department of Psychological Science, University of Texas Rio Grande Valley, Edinburg, TX, USA*

ARTICLE INFO

*Article history:*
Received 2 September 2021
Reviewed 13 December 2021
Revised 13 December 2021
Accepted 16 December 2021
Action editor Roberto Cubelli
Published online xxx

*Keywords:*
Race-norms
Individual comparison standard
Hold measures
Preexisting neuropsychological skill level
African American

ABSTRACT

Groupings of neurologically intact African Americans have mean scores below those for Caucasian Americans on a wide range of neuropsychological tests. This provided the impetus for the creation of race-norms that reduce false positive African American misclassifications (i.e., neurologically intact individuals misdiagnosed with neurocognitive impairment). Race-norming of neuropsychological tests has proved controversial as evidenced by a recent widely publicized lawsuit filed by African American ex-football players against the National Football League (NFL) 2014 concussion settlement that prompted the NFL to halt the use of race-norms and to review past claims for evidence of racial bias. Here it is argued that there is a viable alternative to the use of race-norms in neuropsychological assessment whereby false positives are reduced through the use of obtained postinjury hold measures to estimate preexisting skill levels with standard norms. These performance-based estimates: (a) can be used with all race/ethnic minorities; (b) are applicable to all neuropsychological tests; (c) are more accurate than using the mean of published norms for members of any race/ethnic grouping who possess "true" preexisting neuropsychological skill levels at either end of the normal curve; (d) promote fairness in testing as individuals of differing race/ethnic groupings but otherwise similar demographic profiles, test scores, and estimated preexisting skill levels have the same assessed level of neurocognitive impairment; and (e) the implied causation of lower preexisting estimates in a racially diverse patient is not race per se but behavioral correlates of race like quality of education and acculturation that have demonstrated effects on test scores.

© 2021 Elsevier Ltd. All rights reserved.

With an announcement that made the national television news, the US National Football League (NFL) pledged to halt the use of race-norms in neuropsychological assessments conducted within its concussion settlement agreement of 2014 and to review past claims for evidence of racial bias (Associated Press, 2021). This decision was sparked by a class action lawsuit filed by two African American ex-players (Kevin Henry & Najeh Davenport *vs* National Football League, 2020) that argued the use of race-norms in neuropsychological assessment discriminates against African Americans in violation of equal protection guarantees of the U.S. Constitution. The suit was dismissed, and the two sides were ordered

E-mail address: drgdrg13@yahoo.com.
[1] Department of Psychological Science, University of Texas Rio Grande Valley, 1201 W. University Drive, Edinburg, TX 78541, USA.
https://doi.org/10.1016/j.cortex.2021.12.003
0010-9452/© 2021 Elsevier Ltd. All rights reserved.

Please cite this article as: Gasquoine, P. G., Performance-based alternatives to race-norms in neuropsychological assessment, Cortex, https://doi.org/10.1016/j.cortex.2021.12.003

into arbitration (Dale, 2021). The whole incident provides a unique opportunity for the field of clinical neuropsychology to understand external reactions to one of its most controversial and poorly researched practices, namely its treatment of race, described as a "major weakness" (Possin et al., 2020, p. 377) in a leading American neurology journal.

While the NFL claimed that the use of race-norms in neuropsychological assessment was industry-standard, the frequency of such usage is actually unknown. The purpose of this article is to argue that from a professional standpoint there have been, and are, alternatives to the use of race-norms in neuropsychological assessment. These alternatives that use performance-, as opposed to race-based adjustments, involve individual comparison standard estimates of preexisting skill level.

## 1. The basis of the lawsuit against the NFL

The NFL concussion settlement agreement of 2014 provides monetary compensation for ex-players who have incurred certain delayed-onset, degenerative neurological diagnoses, presumably resulting from sport-related concussions. Two such diagnoses are known as Neurocognitive Impairment Level 1.5 (moderate to severe decline/early or mild dementia) and Level 2 (severe decline/moderate dementia).

The class action lawsuit alleged payouts were being denied for such diagnoses by a formula that explicitly and deliberately discriminates on the basis of race. This formula uses neuropsychological race-norms (Heaton et al., 2004) to determine if obtained ex-player test scores are approximately 1.5*SD* (Level 1 Impairment: not eligible for monetary compensation, but requiring of further monitoring), 1.7–1.8*SD* (Level 1.5), or 2*SD* (Level 2) below preexisting skill levels, estimated at the 50th percentile of the race-norms.

The plaintiffs argued that the use of race-norms meant a Caucasian American ex-player who scored the same as an African American ex-player on $\geq$ 2/5 domains in a non-serial, settlement mandated, standardized neuropsychological test battery was more likely to qualify for compensation, as the Caucasian American's preexisting skill level estimate was automatically assumed to be higher solely on the basis of race. African Americans constitute a majority of the settlement class of over 20,000 ex-players (the proportion of active African American players in the NFL is about 70%).

The plaintiffs were also seeking monetary damages for the pain, suffering, embarrassment, and humiliation they experienced from perceptions that neuropsychological race-norms imply race-based intellectual inferiority/superiority. There were thus two interrelated elements to the lawsuit: (a) that race-norms result in brain injury false negative misclassification error disproportionally affecting African Americans; and (b) that race-norms give the perception of racial bias. Race-norms or other race-based numerical adjustments have previously been outlawed for discriminating against Caucasian Americans in other cognitive (ability) testing arenas, including personnel selection (Gottfredson, 1994) and U.S. undergraduate college admissions (Fisher *vs* University of Texas at Austin, 2016; Gratz et al. *vs* Bollinger et al., 2003). The American Psychological Association (2021a) highlighted various historical missteps in the treatment of race within the broad discipline of psychology, offering an apology to persons of color in the hope of creating a more equitable, diverse, and inclusive association, discipline, and society. The use of race-norms in neuropsychology was not specifically addressed (American Psychological Association, 2021b). Within neuroscience, Cubelli and Della Sala (2018) argued for a ban on use of the term *race*, including within manuscripts submitted to Cortex, in favor of better-defined biological categories and socio-cultural constructs.

## 2. Historical perspectives on African versus Caucasian American differences on cognitive tests

Large *N*, multi-test, race-norming in American neuropsychology dates from Heaton et al. (2004) with the only other such initiate being a Mayo Clinic sponsored project on a convenience sample of African Americans >55 years of age from Jacksonville, Florida (Lucas et al., 2005). Smaller scale normative studies of African American norms for various sets of popular neuropsychological tests have also been published (e.g., Norman et al., 2011). Prior to such publications, race was rarely reported in neuropsychological studies (O'Bryant et al., 2004), but African versus Caucasian American mean score differences have been a major issue in the field of intelligence testing.

That African Americans as a grouping have a mean score below (now about .67*SD*, initially 1*SD*: Nisbett et al., 2012) that of Caucasian Americans on intelligence tests has rarely been challenged, but deliberations regarding the cause of the difference have been highly contentious with several notable flareups. As illustration: (a) arguing that it was genetic (Jensen, 1969), provoked a reaction described as: "Probably no other single article in the history of psychological publications has been subjected to so many niggling and nit-picking commentaries in so brief a time" (Jensen, 1974, p. 467). (b) In Larry P. *vs* Wilson Riles (1979), claims of racial test bias led to a legal ban on the use of intelligence tests in the placement of African American children in special education in California. The ban never spread despite similar lawsuits being contested in other states (e.g., Georgia State Conference of Branches of NAACP *vs* State of Georgia, 1985) and was soon reversed (Crawford et al. *vs* Honig et al., 1994), as there was no alternative objective method for determining special education placement. (c) Publication of *The Bell Curve* (Herrnstein & Murray, 1994) that argued racial differences in IQ were likely due to genetics and difficult to modify via environmental factors prompted the American Psychological Association Board of Scientific Affairs to convene a panel of intelligence experts to determine why racial group differences in testing occurred. They were unable to figure it out (Neisser et al., 1996). It was into this professional milieu that race-norms for neuropsychological tests began to be published.

## 3. Race-norming of neuropsychological tests

Intelligence is highly positively correlated with performance across multiple neuropsychological domains (e.g., Holdnack

et al., 2017) and research has consistently shown that African Americans as a grouping have lower mean scores than Caucasian Americans on a wide range of neuropsychological tests (Werry et al., 2019). Comprehensive sets of race-norms have only ever been published for African Americans, although other American race/ethnic minority groupings have mean neuropsychological test scores below those of Caucasian Americans (e.g., American Indian and Alaskan Natives: Savin et al., 2020).

The justification for the use of African American race-norms was not based on any reasoned theoretical analysis of the effect of race on tests scores but was purely clinical, as evidenced by their recommended use in the assessment of race/ethnic minorities not of African descent (i.e., Hispanic Americans). Race-norms reduce false positive misclassifications (i.e., neurologically intact individuals misdiagnosed with neurocognitive impairment) of African Americans on neuropsychological tests when preexisting skill level is estimated at the 50th percentile of the race-norm (e.g., Norman et al., 2000).

The chief criticism of race-norms is that race is not a categorical variable that produces discrete subcategories that are homogeneous on some measurable distinguishing variable. Historically, the distinguishing variable was thought to be genetic, but research has now demonstrated that commonly used racial categories, are characterized by genetic heterogeneity and lack clearly defined genetic boundaries (Yudell et al., 2016). In practice, racial categories are defined by self-report, using societal conceptions of racial differences, such as those based upon skin color. These self-reported racial categories are heterogeneous on cultural variables known to effect test scores making neuropsychological cross-racial group comparisons difficult to interpret as to causation (Gasquoine, 1999; 2009). They also fail to account for biracial individuals who comprise 15% of the African American populations (U.S. Bureau of the Census, 2021).

There is no a priori reason to assume that skin color has any effect on neuropsychological test scores, but correlates of skin color may. Within neuropsychology the two most widely studied such correlates shown to effect test scores have been quality of education (Manly et al., 2002; O'Bryant et al., 2007; Pedraza, et al., 2012) and acculturation (Kennepohl et al., 2004; Manly et al., 1998). These variables capture within-group cultural heterogeneity, avoiding the use of race as the defining variable.

Dating from the Jim Crow era, African Americans as a grouping have received less quality education than Caucasian Americans. Racial segregation in housing is now officially illegal in the US, but informal race/ethnic enclaves persist. Nationwide about 20% of African Americans live in majority African American neighborhoods where adults are twice as likely to be unemployed or live below the poverty line than adults in Caucasian American neighborhoods (Sauter, 2021). Such enclaves have lower quality schools.

Acculturation measures varying levels of affiliation of race/ethnic minorities, including African Americans, to the dominant culture. It is a complex multifaceted concept (Tan et al., 2021) that in reference to poor test scores can encompass a host of race/ethnic group differences in substantive social exposures like within country migration patterns, ethnic identity, physical health disparities, test-taking attitudes, sociocultural stress, and perceived racial bias that have hypothesized but, as yet, individually unquantified negative effects on neuropsychological test scores (Glymour, & Manly, 2008; Wong et al., 2000).

## 4. Racial bias in cognitive testing

The chief initiative to clinically address African versus Caucasian American intelligence test mean score differences has been the search for a culture-fair intelligence test. This began before World War I with the testing of poorly educated European immigrants on Ellis Island (Boake, 2002) and has primarily involved the use of visual-perceptual and visual-motor processing tasks and various attempts to define racial test bias. Although multiple definitions have been proposed (e.g., Flaugher, 1978; Schmidt & Hunter, 1974; Steele, 1997; Williams, 1972), no definition of racial test bias has proved sufficiently accurate to allow construction of a true culture-fair intelligence test (i.e., one that would show no difference in mean scores between African and Caucasian American groupings).

One modern conception of racial test bias is stereotype threat, namely the concept that test performance can confirm racial stereotypes, that has been extensively researched in the social psychology literature. It was originally introduced to explain poor achievement test performance by African versus Caucasian Americans (Steele, 1997), but has since been expanded to explain multiple, non-racial, experimental, group behavioral differences including poor neuropsychological test performance of college students with remote histories of concussion (Suhr & Gunstad, 2005). Meta-analytical estimation of the real-world effect of stereotype threat on intelligence test performance is small (<.2*SD*: Walton & Spencer, 2009), the mechanism of operation is unclear (Pennington et al., 2016), and various interventions have been shown to successfully mitigate its effect (Liu et al., 2021).

### 4.1. *Racial bias stress*

Numerous studies have investigated the relationship between mental health and racial bias, viewing the latter as a psychosocial stressor. In these studies, racial bias is measured by self-report checklists like the *Everyday Discrimination Scale* (Williams et al., 1997) that measure perceived (as opposed to actual) racial bias. Perceived racial bias is not unique to African Americans and is reported at differing frequencies within all U.S. race/ethnic groupings including Caucasian Americans (Bower et al., 2013). In racial bias studies mental health has primarily been measured as non-specific symptoms of, for example, depression, obsessive-compulsion, or somatization, or as subjective well-being, self-esteem, or life-satisfaction.

Most studies are correlational, with meta-analysis giving an effect size between perceived racial bias and mental health of −.23 (Paradies et al., 2015). The real-world effect of this

psychosocial stress model is unclear as government-sponsored, national surveys of community dwelling adults have paradoxically found that groupings of African, Asian, and Hispanic Americans have lower frequencies of common (mood, anxiety, impulse, and substance use) DSM-defined mental disorders than non-Hispanic Caucasian Americans (Grant et al., 2015; Hasin & Grant, 2015; Vilsaint et al., 2019). Psychosocial stress load (i.e., the sum of all stressors), has never been considered to effect neuropsychological test performance to the point where it would be included in normative studies (Silveira et al., 2020). Possin et al. (2020) argued for a regression-based normative system for neuropsychological tests that adjusts for social determinants of brain health, consisting of multiple variables including psychosocial stress (perceived discrimination, early-life adversity, stereotype threat). This approach would prove difficult to implement in practice requiring a whole new normative system for every test and an extremely large sample to accommodate all the variables. As a first step it would be necessary to demonstrate that these variables are able to be operationalized and actually do have a demonstratable, and not just a hypothesized effect, on neuropsychological test scores.

## 5. Performance-based alternatives to race-norms

As with race-norms, false positives in non-serial neuropsychological evaluations of race/ethnically diverse patients can be reduced through the use of performance-based individual comparison standards to estimate preexisting skill level (Gasquoine, 2009). These approaches require only a single normative table (stratified by age, education, and/or gender), ideally constructed from a national sample with proportional representation of race/ethnic groupings based on census data. The estimated preexisting skill level applied to this normative table is not fixed at the 50th percentile but varies from case-to-case based upon obtained postinjury scores from tests measuring skills that are relatively resistant to the particular type of brain injury suspected, known as *hold* measures.

The use of hold measures to estimate preexisting skill level is not specific to race/ethnic assessments and dates from Babcock (1930). As with other forms of test selection in clinical neuropsychology, subsequent research conducted predominantly outside the race/ethnic arena has not reached any consensus as to a best hold measure to use for any given form of brain injury. Popular choices include Wechsler intelligence test Vocabulary, Information, or Matrix Reasoning subtest score percentiles (Holdnack et al., 2013), or score percentiles from phonetically irregular word pronunciation measures (Blair & Spreen, 1989; Del Pino et al., 2018; Wechsler, 2001; Wilkinson & Robertson, 2017). Some standard hold measures are counter mandated with certain forms of suspected brain injury, as with Matrix Reasoning scores in a patient suspected to have non-dominant hemisphere compromise.

Hold scores have also been combined with demographic data into multiple regression predictor models. Some of these models, such as that of the Test of Premorbid Functioning (The Psychological Corporation, 2009) incorporate race-based adjustments, also a subject of the lawsuit against the NFL. That particular model has proved procedurally difficult for clinicians to correctly implement (Shura et al., 2020). Preexisting skill level has also been estimated from historical test data like school achievement records, an approach of questionable validity when schools "teach to the test".

Consensus has not been reached as studies to determine the best measure to estimate preexisting neuropsychological skill level with various brain injured patient groupings are confounded by the lack of "true" preexisting skill levels to use as validation criteria. Thus, studies employ proxy comparisons, such as predicting actual Full-Scale IQ in neurologically intact samples (Bright & van der Linde, 2020), or postinjury Full-Scale IQ in traumatic brain injury samples (Joseph et al., 2019). In such studies, all methods of estimating preexisting skill levels from hold measures show regression to the mean in cases where the "true" estimate is at either end of the normal distribution (e.g., Bright et al., 2016). This implies they overestimate preexisting skill level in patients with very low postinjury scores and underestimate it for those with very high postinjury scores. This statistical phenomenon is common to all distributions of extreme scores.

Advantages of performance-based methods to estimate preexisting level include:

- They can be used with all race/ethnic minorities.
- They are applicable to all neuropsychological tests with the proviso that it is an assumption that performance in the domain of the estimate (e.g., language) is highly correlated with performance across other domains (e.g., executive). Correlations among neuropsychological measures average around .5 in the general population (Crawford et al., 2007).
- They are more accurate than using the mean of published norms for members of any race/ethnic grouping who possess "true" preexisting neuropsychological skill levels at either end of the normal curve. For example, if a Caucasian American's preexisting skill estimate is at the 16th percentile, the application of a 1*SD* criterion cutoff defines neurocognitive impairment as scores <2nd percentile. In contrast, if the estimated preexisting skill level is the normative mean, the same criterion cutoff defines neurocognitive impairment as scores <16th percentile, increasing the likelihood of false positive misclassification.
- From a fairness (absence of bias) in testing perspective (American Educational Research Association, American Psychological Association, National Council on Measurement in Education, 2014), individuals of differing race/ethnic groupings but otherwise similar demographic profiles, test scores, and estimated preexisting skill levels have the same assessed level of neurocognitive impairment, as scores are analyzed using the same normative table.
- The implied causation of lower preexisting estimates in a racially diverse patient is not race per se but behavioral correlates of race like quality of education and acculturation that have demonstrated effects on test scores. These correlates are assumed to affect not only the actual neuropsychological test scores but also the performance-based estimates of preexisting skill level.

Please cite this article as: Gasquoine, P. G., Performance-based alternatives to race-norms in neuropsychological assessment, Cortex, https://doi.org/10.1016/j.cortex.2021.12.003

## 6. Effect of race-norms and performance-based preexisting estimates on false positive and false negative misclassification rates

Error in non-serial neuropsychological assessments to determine the presence of neurocognitive impairment in adults is inevitable to some extent (Reitan & Wolfson, 2008). In addition to random testing error and non-brain injury-related patient situational variables (e.g., drug effects), the major contributor to this error is that preexisting performance on tests must be estimated either as the 50th percentile of published norms or by an individual performance-based comparison standard. Total neuropsychological testing error is measured in research studies as false positive (or specificity = 1 - %false positive) and false negative (or sensitivity = 1 - %false negative) misclassification rates. In group studies, the two misclassification rates are negatively related, such that any decrease in one misclassification rate will likely increase the misclassification rate of the other.

When estimating the preexisting skill level at the 50th percentile, any adjustment that lowers this estimate (the mean) will automatically lower the group false positive rate in studies of neurologically intact samples. For example, race-norms lowered the African American false positive rate on the California Verbal Learning Test (CVLT) Trials 1–5 from 36% to 18% (Norman et al., 2000). Similarly, using performance-based preexisting estimates that predominantly, within a given sample, fall below the 50th percentile will also reduce the false positive rate. For example, from 41% to 11% for the same CVLT-II measure among a grouping of neurologically intact, bilingual, Hispanic Americans tested in Spanish using preexisting skill level estimates from Spanish language vocabulary scores (Gasquoine & Gonzalez, 2012). In both these studies the expected false positive rate based on the normal distribution was 16% (i.e., a $\leq$ 1*SD* criterion cutoff). There have been no studies that have directly compared false positive error rates from the two approaches.

False negative misclassification rates cannot be estimated from normative or other studies that test only neurologically intact participants. As with false positive rates they vary by measure, but they also vary by the type of brain injury suspected. Determining false negative rates within the NFL settlement class is confounded by an inability to validate the presence of concussive brain injury independently of a neuropsychological assessment, especially given that the contribution of sport-related concussion to delayed-onset dementia remains contentious (Barr, 2020; LoBue & Cullum, 2020). The modern variant of sport-related traumatic encephalopathy that has recently become associated with American football participation can only definitively be diagnosed at autopsy (McKee et al., 2016), where the uniqueness of its neuropathology in comparison to that of other neurogenerative diseases has been questioned (e.g., Randolph, 2018). Thus, the effect of using race-norms or individual performance-based preexisting level estimates on false negative rates in the neuropsychological assessment of ex-NFL players is unknown.

Increasing false negative rates has been a theoretical criticism of race-norming (Brandt, 2007; Manly 2005), but it is an inevitable side-effect of decreasing false positive rates. Both types of misclassification errors are bad and in practice a clinical neuropsychologist must strike a balance between the probability of the two types of misclassification errors in any given case. For example, in an independent practice assessment where the suspected brain injury is mild the clinical neuropsychologist may emphasize avoiding a false positive misclassification error. This may not be so defensible in a forensic setting.

## 7. Conclusion and future directions

Race-norms were introduced into neuropsychology as groupings of African Americans have lower mean scores than Caucasian Americans on a wide range of tests. They reduce false positive misclassification rates in neurologically intact African American samples. Reducing false positive misclassification rates increases false negative misclassification rates, although the size of the increase cannot be determined from studies that use neurologically intact samples.

Race-norms assume that subcategories of race form homogeneous groupings, an assumption that is incorrect. Commonly used racial categories, defined by self-report are genetically indistinct and culturally heterogeneous on variables known to impact neuropsychological test scores. The use of race-norms thus contributes to perceptions of racial stereotyping as evidenced by the recent lawsuit against the NFL.

Performance-based estimates of preexisting levels from obtained postinjury hold measures can also reduce false positive misclassifications in race/ethnic minorities and they do so without considering race as a variable. Race/ethnic differences in scores on these measures are conceived as reflecting variations in the interaction between genetics and correlates of race such as quality of education and/or acculturation. Most importantly, the use of individual performance-based preexisting estimates are viewed as promoting fairness in testing as individuals of differing race/ethnic groupings but otherwise similar demographic profiles, test scores, and estimated preexisting skill levels are assigned the same level of neurocognitive impairment.

Given the Flynn effect, the two comprehensive sets of published race-norms are now outdated. The Heaton et al. (2004) race-norms include scores from the original Wechsler Adult Intelligence Scale (Wechsler, 1955) and its first revision (WAIS-R: Wechsler, 1981). The Mayo Clinic race-norms are linked with the WAIS-R. Adoption of performance-based individual comparison standard estimates of preexisting skill level in the determination of neurocognitive impairment in adults can eliminate the laborious process of creating updated sets of race-norms, while abolishing the perception that skin color affects performance on cognitive tests.

There is much to be learned about the use of performance-based estimates of preexisting skill levels with U.S. race/ethnic minorities as it has attracted very little research attention. Fundamental questions include: (a) what hold measures work best with certain race/ethnically or culturally defined groupings (e.g., Spanish-English bilinguals; foreign-born) (b) are hold measures in specific domains (e.g., language); predictive of

Please cite this article as: Gasquoine, P. G., Performance-based alternatives to race-norms in neuropsychological assessment, Cortex, https://doi.org/10.1016/j.cortex.2021.12.003

performance across all other domains (e.g., fine motor coordination); and (c) what false negative rates occur with a given estimate within known brain injury groupings. Much of this research could potentially involve retrospective analysis of existing patient series.

## Declaration of competing interest

The author declares no conflict of interest.

## Acknowledgement

This study was not funded.

## REFERENCES

American Educational Research Association, American Psychological Association, National Council on Measurement in Education. (2014). Washington, DC: Author.

American Psychological Association. (2021). Apology to people of color for APA's role in promoting, perpetuating, and failing to challenge racism, racial discrimination, and human hierarchy in the U.S.. *Retrieved from Apology to People of Color for APA's Role in Promoting, Perpetuating, and Failing to Challenge Racism, Racial Discrimination, and Human Hierarchy in U.S.*.

American Psychological Association. (2021). Historical chronology: examining psychology's contributions to the belief in racial hierarchy and perpetuation of inequality for people of color in the U.S.. *Retrieved from Historical chronology (apa.org)*.

Associated Press. (2021). *NFL to halt 'race-norming,' review Black claims in $1 billion concussion settlement*. Retrieved from NFL to halt 'race-norming,' review Black claims in $1 billion concussion settlement espn.com.

Babcock, H. (1930). An experiment in the measure of mental deterioration. *Archives of Psychology, 117*, 1–105.

Barr, W. B. (2020). Links between traumatic brain injury and dementia remain poorly defined. *Archives of Clinical Neuropsychology, 35*, 128–132. https://doi.org/10.1093/arclin/acz075

Blair, J. R., & Spreen, O. (1989). Predicting Premorbid IQ: A Revision of the National Adult Reading Test. *The Clinical Neuropsychologist, 3*, 129–136. https://doi.org/10.1080/13854048908403285

Boake, C. (2002). From the Binet-Simon to the Wechsler-Bellevue: Tracing the history of intelligence testing. *Journal of Clinical and Experimental Neuropsychology, 24*, 383–405. https://doi.org/10.1076/jcen.24.3.383.981

Bower, K. M., Thorpe, R. J., Jr., & LaVeist, T. A. (2013). Perceived racial discrimination and mental health on low-income, urban-dwelling whites. *International Journal of Health Services, 43*, 267–280. https://doi.org/10.2190/HS.43.2.e

Brandt, J. (2007). 2005 INS Presidential address: Neuropsychological crimes and misdemeanors. *The Clinical Neuropsychologist, 21*, 553–568. https://doi.org/10.1080/13854040600770800

Bright, P., Hale, E., Gooch, V. J., Myhill, T. D., & van der Linde, I. (2016). The National Adult Reading Test: Restandardisation Against the Wechsler Adult Intelligence Scale – Fourth Edition. *Neuropsychological Rehabilitation*. https://doi.org/10.1080/09602011.2016.1231121

Bright, P., & van der Linde, I. (2020). Comparison of methods for estimating premorbid intelligence. *Neuropsychological Rehabilitation, 30*, 1–14. https://doi.org/10.1080/09602011.2018.1445650

*Crawford et al. vs. Honig et al., 37 F.3d 485.* (1994). U.S. 9th Cir.

Crawford, J. R., Garthwaite, P. H., & Gault, C. B. (2007). Estimating the percentage of the population with abnormally low scores (or abnormally large score differences) on standardized neuropsychological test batteries: A generic method with applications. *Neuropsychology, 21*, 419–430. https://doi.org/10.1037/0894-4105.21.4.419

Cubelli, R., & Della Salla, S. (2018). Taking race out of neuroscience too. *Cortex, 99*, A1–A2.

Dale, M. (2021). *Federal judge tosses suit over 'race-norming' in NFL dementia tests*. Retrieved from: Federal judge tosses suit over 'race-norming' in NFL dementia tests inquirer.com.

Del Pino, R., Peña, J., Ibarretxe-Bilbao, N., Schretlen, D. J., & Ojeda, N. (2018). Demographically Calibrated Norms for Two Premorbid Intelligence Measures: The Word Accentuation Test and Pseudo-words Reading Subtest. *Frontiers in Psychology, 9*. https://doi.org/10.3389/fpsyg.2018.01950

Fisher vs University of Texas at Austin. (2016). *579 U.S.*

Flaugher, R. L. (1978). The many definitions of test bias. *American Psychologist, 33*, 671–679. https://doi.org/10.1037/0003-066X.33.7.671

Gasquoine, P. G. (1999). Variables moderating cultural and ethnic differences in neuropsychological assessment: The case of Hispanic Americans. *The Clinical Neuropsychologist, 13*, 376–383. https://doi.org/10.1076/clin.13.3.376.1735

Gasquoine, P. G. (2009). Race-norming of neuropsychological tests. *Neuropsychology Review, 19*, 250–262. https://doi.org/10.1007/s11065-009-9090-5

Gasquoine, P. G., & Gonzalez, C. D. (2012). Using monolingual neuropsychological test norms with bilingual Hispanic Americans: Application of an individual comparison standard. *Archives of Clinical Neuropsychology, 27*, 268–276. https://doi.org/10.1093/arclin/acs004

*Georgia State Conference of Branches of NAACP vs State of Georgia, 775 F.21403.*(1985).

Glymour, M. M., & Manly, J. J. (2008). Lifecourse social conditions and racial and ethnic patterns of cognitive aging. *Neuropsychology Review, 18*, 223–254. https://doi.org/10.1007/s11065-008-9064-z

Gottfredson, L. S. (1994). The science and politics of race-norming. *American Psychologist, 49*, 955–963. https://doi.org/10.1037/0003-066X.49.11.955

Grant, B. F., Goldstein, R. B., Saha, T. D., Chou, P. C., Jung, J., Zhang, H., Pickering, R. P., Ruan, W. J., Smith, S. M., Huang, B., & Hasin, D. S. (2015). Epidemiology of DSM-5 Alcohol Use Disorder: Results from the National Epidemiologic Survey on Alcohol and Related Conditions-III. *JAMA Psychiatry, 72*, 757–766. https://doi.org/10.1001/jamapsychiatry.2015.0584

*Gratz et al. vs Bollinger et al., 539 U.S. 244.* (2003).

Hasin, D. S., & Grant, B. F. (2015). The National Epidemiological Survey on Alcohol and Related Conditions (NESARC) Waves 1 and 2: Review and Summary of Findings. *Social Psychiatry and Psychiatric Epidemiology, 50*, 1609–1640. https://doi.org/10.1007/s00127-015-1088-0

Heaton, R. K., Miller, S. W., Taylor, M. J., & Grant, I. (2004). *Revised comprehensive norms for an expanded Halstead-Reitan battery: Demographically adjusted neuropsychological norms for African American and Caucasian Adults*. Odessa, FL: Psychological Assessment Resources.

Herrnstein, R. J., & Murray, C. (1994). *The bell curve: Intelligence and class structure in American life*. NY: Free Press.

Holdnack, J. A., Schoenberg, M. R., Lange, R. T., & Iverson, G. L. (2013). Predicting premorbid ability for WAIS-IV, WMS-IV, and WASI-II. In J. A. Holdnack, L. W. Drozdick, L. G. Weiss, &

G. L. Iverson (Eds.), *WAIS-IV, WMS-IV, and ACS: Advanced clinical interpretation* (pp. 217–278). San Diego, CA: Academic Press.

Holdnack, J., Tulsky, D., Brooks, B., Slotkin, J., Gershon, R., Heinemann, A., & Iverson, G. (2017). Interpreting patterns of low scores on the NIH toolbox cognition battery. *Archives of Clinical Neuropsychology, 32*, 574–584. https://doi.org/10.1093/arclin/acx032

Jensen, A. R. (1969). How much can we boost IQ and scholastic achievement? *Harvard Educational Review, 39*, 1–123. https://doi.org/10.17763/haer.39.1.l3u15956627424k7

Jensen, A. R. (1974). The strange case of Dr. Jensen and Mr. Hyde. *American Psychologist, 29*, 467–468. https://doi.org/10.1037/h0038149

Joseph, A.-L. C., Lippa, S. M., McNally, S. M., Garcia, K. M., Leary, J. B., Dsurney, J., & Chan, L. (2019). Estimating premorbid intelligence in persons with traumatic brain injury: An examination of the Test of Premorbid Functioning. *Applied Neuropsychology: Adult*. https://doi.org/10.1080/23279095.2019.1661247

Kennepohl, S., Shore, D., Nabors, N., & Hanks, R. (2004). African American acculturation and neuropsychological test performance following traumatic brain injury. *Journal of the International Neuropsychological Society, 10*, 566–577. https://10.1017/S1355617704104128.

*Kevin Henry and Najeh Davenport* vs *National football League*.(2020). Retrieved from: https://www.classaction.org/media/henry-et-al-v-national-football-league-et-al.pdf.

*Larry P.* vs *Wilson Riles, 495 F. Supp 926 (N. D. Cal)*. (1979).

Liu, S., Liu, P., Wong, M., & Zhang, B. (2021). Effectiveness of stereotype threat interventions: A meta-analytic review. *Journal of Applied Psychology, 106*, 921–949. https://doi.org/10.1037/apl0000770

LoBue, C., & Cullum, C. M. (2020). Beyond the headlines: The actual evidence that traumatic brain injury is a risk factor for later-in-life dementia. *Archives of Clinical Neuropsychology, 35*, 123–127. https://doi.org/10.1093/arclin/acz074

Lucas, J. A., Ivnik, R. J., Willis, F. B., Ferman, T. J., Smith, G. E., Parfitt, F. C., Petersen, R. C., & Graff-Radford, N. R. (2005). Mayo's older African Americans normative studies: Normative data for commonly used clinical neuropsychological measures. *The Clinical Neuropsychologist, 19*, 162–183. https://doi.org/10.1080/13854040590945265

Manly, J. J. (2005). Advantages and disadvantages of separate norms for African Americans. *The Clinical Neuropsychologist, 19*, 270–275. https://doi.org/10.1080/13854040590945346

Manly, J. J., Jacobs, D. M., Touradji, P., & Small, S. A. (2002). Reading level attenuates differences in neuropsychological test performance between African American and White elders. *Journal of the International Neuropsychological Society, 8*, 341–348. https://doi.org/10.10170S1355617705051040.

Manly, J. J., Miller, S. W., Heaton, R. K., Byrd, D., Reilly, J., Velasquez, R. J., Saccuzzo, D. P., & Grant, I. (1998). The effect of African-American acculturation on neuropsychological test performance in normal and HIV-positive individuals. *Journal of the International Neuropsychological Society, 4*, 291–302. https://doi.org/10.1017/S1355617798002914

McKee, A. C., Cairns, N. J., Dickson, D. W., Folkerth, R. D., Keene, C. D., Litvan, I., Perl, D. P., Stein, T. D., Vonsattel, J.-P., Stewart, W., Tripodis, Y., Crary, J. F., Bieniek, K. F., Dams-O'Connor, K., Alvarez, V. E., & Gordon, W. A. (2016). The first NINDS/NIBIB consensus meeting to define neuropathological criteria for the diagnosis of chronic traumatic encephalopathy. *Acta Neuropathologica, 131*, 75–86. https://doi.org/10.1007/s00401-015-1515-z

Neisser, U., Boodoo, G., Bouchard, T. J., Boykin, A. W., Brody, N., Ceci, S. J., Halpern, D. F., Loehlin, J. C., Perloff, R., Sternberg, R. J., & Urbina, S. (1996). Intelligence: Knowns and unknowns. *American Psychologist, 51*, 77–100. https://doi.org/10.1037/0003-066X.51.2.77

Nisbett, R. E., Aronson, J., Blair, C., Dickens, W., Flynn, J., Halpern, D. F., & Turkheimer, E. (2012). Intelligence: New findings and theoretical developments. *American Psychologist, 67*, 130–159. https://doi.org/10.1037/a0026699

Norman, M. A., Evan, J. D., Miller, S. W., & Heaton, R. K. (2000). Demographically Corrected Norms for the California Verbal Learning Test. *Journal of Clinical and Experimental Neuropsychology, 22*, 80–94. https://doi.org/10.1076/1380-3395(200002)22:1;1-8;FT080

Norman, M. A., Moore, D. J., Taylor, M., Franklin, D., Jr., Cysique, L., Ake, C., Lazarretto, D., Vaida, F., & Heaton, R. K. (2011). Demographically Corrected Norms for African Americans and Caucasians on the Hopkins Verbal Learning Test–revised, Brief Visuospatial Memory Test–revised, Stroop Color and Word Test, and Wisconsin Card Sorting Test 64-card Version. *Journal of Clinical and Experimental Neuropsychology, 33*, 797–804. https://doi.org/10.1080/13803395.2011.559157

O'Bryant, S. E., Lucas, J. A., Willis, F. B., Smith, G. E., Graff-Radford, N. R., & Ivnik, R. J. (2007). Discrepancies between self-reported years of education and estimated reading level among elderly community-dwelling African Americans: Analysis of the MOAANS data. *Archives of Clinical Neuropsychology, 22*, 327–332. https://doi.org/10.1016/j.acn.2007.01.007

O'Bryant, S. E., O'Jile, J. R., & McCaffrey, R. J. (2004). Reporting of demographic variables in neuropsychological research: Trends in the current literature. *The Clinical Neuropsychologist, 18*, 229–233. https://doi.org/10.1080/13854040490501439

Paradies, Y., Ben, J., Denson, N., Elias, A., Priest, N., Pieterse, A., Gupta, A., Kelaher, M., & Gee, G. (2015). Racism as a determinant of health: A systematic review and meta-analysis. *Plos One, 10*, Article e0138511. https://doi.org/10.1371/journal.pone.0138511

Pedraza, O., Clark, J. A., O'Bryant, S. E., Smith, G. E., Ivnik, R. J., Graff-Radford, N. R., Willis, F. B., Petersen, R. C., & Lucas, J. A. (2012). Diagnostic validity of age and education corrections for the Mini-Mental State Examination in older African Americans. *Journal of the American Geriatrics Society, 60*, 328–331. https://doi.org/10.1111/j.1532-5415.2011.03766.x

Pennington, C. R., Heim, D., Levy, A. R., & Larkin, D. T. (2016). Twenty years of stereotype threat research: A review of psychological mediators. *Plos One, 11*, Article e0146487. https://doi.org/10.1371/journal.pone.0146487

Possin, K. L., Tsoy, E., & Windon, C. C. (2020). Perils of race-based norms in cognitive testing: The case of former NFL players. *JAMA Neurology*. https://doi.org/10.1001/jamaneurol.2020.4763

Randolph, C. (2018). Chronic traumatic encephalopathy is not a real disease. *Archives of Clinical Neuropsychology, 33*, 644–648. https://doi.org/10.1093/arclin/acy063

Reitan, R. M., & Wolfson, D. (2008). Can neuropsychological testing produce unequivocal evidence of brain damage? I. Testing for specific deficits. *Applied Neuropsychology, 15*, 33–38. https://doi.org/10.1080/09084280801917350

Sauter, M. B. (2021). *The most segregated cities in America. Retrieved from: The Most Segregated Cities in America – 24/7 Wall St.* 247wallst.com.

Savin, M., Summers, A., Crook, C., Aghvinian, M., Byrd, D., Armenta, R., Franklin, D., Marcotte, T., & Rivera Mindt, M. (2020). Current norms may overestimate rates of neurocognitive impairment among American Indian and Alaskan Native adults. *Archives of Clinical Neuropsychology, 35*, 779. https://doi.org/10.1093/arclin/acaa067.06

Schmidt, F. L., & Hunter, J. E. (1974). Racial and ethnic bias in psychological tests: Divergent implications of two definitions of test bias. *American Psychologist, 29*, 1–13. https://doi.org/10.1037/h0037806

Shura, R. D., Ord, A. S., Martindale, L., Miskey, H. M., & Taber, K. H. (2020). Test of Premorbid Functioning: You're doing it wrong, but does it matter? *Archives of Clinical Neuropsychology*. https://doi.org/10.1093/arclin/acaa025

Silveira, K., Garcia-Barrera, M. A., & Smart, C. M. (2020). Neuropsychological impact of trauma-related mental illnesses : A systematic review of clinically meaningful results. *Neuropsychology Review, 30*, 310–344. https://doi.org/10.1007/s11065-020-09444-6

Steele, C. M. (1997). A threat in the air: How stereotypes shape intellectual identity and performance. *American Psychologist, 52*, 613–629. https://doi.org/10.1037//0003-066x.52.6.613

Suhr, J. A., & Gunstad, J. (2005). Further exploration of the effect of "diagnostic threat" on cognitive performance in individuals with mild head injury. *Journal of the International Neuropsychological Society, 11*, 23–29. https://doi.org/10.10170S1355617705050010.

Tan, Y. W., Burgess, G. H., & Green, R. J. (2021). The effects of acculturation on neuropsychological test performance; A systematic literature review. *The Clinical Neuropsychologist, 35*, 541–571. https://doi.org/10.1080/13854046.2020.1714740

The Psychological Corporation. (2009). *Test of premorbid functioning: Advanced clinical solutions for use with the WAIS-IV and WMS-IV*. San Antonio, TX: Author.

U.S. Bureau of the Census. (2021). *Race and ethnicity in the United States: 2010 census and 2020 census*. Retrieved from Race and Ethnicity in the United States: 2010 Census and 2020 Census.

Vilsaint, C. L., NeMoyer, A., Fillbrunn, M., Sadikova, E., Kessler, R. C., Sampson, N. A., Alvarez, K., Greif Green, J., McLaughlin, K. A., Chen, R., Williams, D. R., sJackson, J. S., & Alegría, M. (2019). Racial/ethnic differences in 12-month prevalence and persistence of mood, anxiety, and substance use disorders: Variation by nativity and socioeconomic status. *Comprehensive Psychiatry, 89*, 52–60. https://doi.org/10.1016/j.comppsych.2018.12.008

Walton, G. M., & Spencer, S. J. (2009). Latent ability: Grades and test scores systematically underestimate the intellectual ability of negatively stereotyped students. *Psychological Science, 20*, 1132–1139. https://doi.org/10.1111/j.1467-9280.2009.02417.x

Wechsler, D. (1955). *Wechsler Adult Intelligence Scale*. San Antonio, TX: The Psychological Corporation.

Wechsler, D. (1981). *Wechsler Adult Intelligence Scale-Revised*. San Antonio, TX: The Psychological Corporation.

Wechsler, D. (2001). *Wechsler Test of Adult Reading: WTAR*. San Antonio, TX: The Psychological Corporation.

Werry, A. E., Daniel, M., & Bergström, B. (2019). Group differences in normal neuropsychological test performance for older non-Hispanic White and Black/African American adults. *Neuropsychology, 33*, 1089–1100. https://doi.org/10.1037/neu0000579

Wilkinson, G. S., & Robertson, G. J. (2017). *Wide Range Achievement Test* (5th ed.). Boston, MA: Pearson.

Williams, R. L. (1972). *The Black Intelligence Test of Cultural Homogeneity*. St. Louis, MO: Williams & Associates.

Williams, D. R., Yu, Y., Jackson, J., & Anderson, N. (1997). Racial differences in physical and mental health: Socioeconomic status, stress, and discrimination. *Journal of Health Psychology, 2*, 335–351. https://doi.org/10.1177/135910539700200305

Wong, T. M., Strickland, T. L., Fletcher-Jansen, E., Ardila, A., & Reynolds, C. R. (2000). Theoretical and practical issues in the neuropsychological assessment and treatment of culturally dissimilar patients. In E. Fletcher-Janzen, T. L. Strickland, & C. R. Reynolds (Eds.), *Critical issues in neuropsychology. Handbook of cross-cultural neuropsychology* (pp. 3–18). NY: Kluwer Academic Publishers.

Yudell, M., Roberts, D., DeSalle, R., & Tishkoff, S. (2016). Taking race out of human genetics. *Science, 351*, 564–565. https://doi.org/10.1126/science.aac4951

Please cite this article as: Gasquoine, P. G., Performance-based alternatives to race-norms in neuropsychological assessment, Cortex, https://doi.org/10.1016/j.cortex.2021.12.003